

Komodo Library System

Project Idea Definition and Detailed Explanation

Problem Description

Traditional library management in many Kenyan secondary schools relies heavily on manual **ledgers**. This manual approach is plagued by several issues:

1. High "search time" for students.
2. Frequent data entry errors
3. The inability to track overdue books effectively, which leads to significant financial loss for the institution.

Kenyan Environment and Context: The Kenyan education sector is currently undergoing a massive shift with the implementation of the Competency-Based Curriculum (CBC). This transition has introduced a wider variety of learning materials and reference books.

In schools across Nairobi, libraries are often understaffed, with teachers doubling as librarians. Managing hundreds of students and thousands of books using paper records is no longer sustainable or accurate.

This system directly contributes to **SDG #4** by ensuring that all students have **equitable access to organized learning materials**. By digitizing the library, the school provides a structured environment for academic excellence.

Furthermore, it supports **Target 4.4** by increasing the number of youth who have relevant ICT skills through daily interaction with a digital database system.

Scope & Objectives

So, What can the system do?

The Digital Library Management System (DLMS) is designed to serve as a comprehensive administrative backbone for school libraries.

It will automate the entire lifecycle of a physical book, from its initial registration in the inventory to its categorization by genre and author. The system will manage the digital registration of all student and staff members, track active loans, calculate overdue fines based on set daily rates, and maintain a queuing system for book reservations. It will also provide reporting tools for librarians to audit the total value of school assets.

And what can the system not do?

The system is strictly a management tool and not a content delivery platform. It will not host, stream, or allow the downloading of digital e-book files or multimedia content. Furthermore, the system will not handle direct financial transactions; while it will calculate and record fines, it will not process external payments via mobile money (like M-Pesa), credit cards, or bank APIs.

A Few Core Objectives of this System Include;

1. **Efficiency:** To reduce the time taken to check out a book from minutes to seconds.
2. **Inventory Integrity:** To ensure 100% accuracy in tracking the status (Available, Loaned, or Reserved) of every book copy.
3. **Sustainability:** To implement an automated fine system that encourages timely returns and provides funds for library maintenance.

Stakeholders of this System include;

- **Students:** They use the system to search for books and view their borrowing history. The system helps them access materials quickly to improve their learning outcomes.
- **Librarians:** They perform the daily operations like issuing books and updating the catalog. The system reduces their administrative burden significantly.
- **School Administration:** They review reports on library usage and inventory value to make informed budget decisions regarding the purchase of new resources.

Main Operations Supported

- **Data Storage:** Managing extensive records for books, authors, publishers, and school members.
- **Transaction Management:** Recording and updating the status of loans, returns, and reservations in real-time.
- **Automated Calculation:** Computing overdue days and corresponding fine amounts without manual intervention.
- **Information Retrieval:** Providing a searchable interface for students to find materials by title, author, or genre.